

Chapter 2: Variables & Data Types

Variables are containers for storing data values. In Python, a variable is created the moment you first assign a value to it using the assignment operator (=). Standard Built-in Data Types in Python: -

Numeric Types: int (whole numbers like 42), float (decimal numbers like 3.14). - Text Sequence Type: str (strings enclosed in single or double quotes, e.g., 'Hello Python'). - Boolean Type: bool (represents truth values True or False). - Type Conversion: You can explicitly convert between types using built-in constructors like int(), float(), and str(). Example variable assignments: x = 10 # Integer pi = 3.14159 # Float name = "AI Bot" # String is_active = True # Boolean